Report Nº 4
Team Out of Lane
Date 21.04.2025

1 Introduction

This report details the activities carried out to date in the 4th sprint. These activities emphasize lane detection and lane keeping, identification of signs and objects and their corresponding reactions to them, as well as the path planning algorithm. These advances are reflected in the classification video and in the 4th report video.

2 Planned activities

Category	*Activity*	*Responsible*
Hardware improvements	Mechanical calibration	Alexander P.
	Research hardware additions (headlight)	Alexander P., Carolina N. and Evelyn C.
	Define other necessary sensors, define use-case, integration (IMU, distance), preprocessing, noise cancelling	Alexander P., Carolina N., Evelyn C. and Pablo F.
	Testing sensors and actuators for proper capture, processing and response.	Alexander P., Carolina N., Evelyn C. and Pablo F.
Sensing and input package	Define use-cases and test given servers information (localization on map, cars interaction, GPS interaction, interaction with objects)	Alexander P., Carolina N., Evelyn C. and Pablo F.
	Environmental server interaction	Alexander P., Carolina N., Evelyn C. and Pablo F.
Computer vision package (RPi)	Implement lane detection	Alexander P., Carolina N. and Pablo F.
	Intersection detection	Alexander P., Carolina N, Evelyn C. and Pablo F.
	Implement traffic sign detection	Carolina N., Evelyn C. and Pablo F.
	Implement traffic light detection	Evelyn C.
	Object detection & classification	Evelyn C.
Project Documentation	Architecture revision	Pablo F.
	Architecture diagram	Evelyn C. and Pablo F.
	Prepare traffic sign dataset	Carolina N. and Evelyn C.
	Define decision making (priorities of actions and state flow)	Alexander P., Carolina N., Evelyn C. and Pablo F.
	Define robustness and safety measures	Alexander P., Carolina N, Evelyn C. and Pablo F.
	Document the 4th report	Carolina N. and Pablo F.

3 Status of planned activities

Activity	*%*	*Observations*
Mechanical calibration	100%	We fixed the small deviation.
Research hardware additions (headlight)	90%	The ultrasonic sensors are installed. The lights are almost ready; final tests are missing.
Define other necessary sensors, define use-case, integration, preprocessing, noise cancelling	80%	We started using the IMU data and added an opto-isolator to calculate the position of the car based on its trajectory. There is still room for improvement in the error handling. *(*)*
Testing sensors and actuators for proper capture, processing and response.	50%	We have tested sensors and actuators.
Define use-cases and test given servers information	10%	We began investigating the data that will be used and potential interactions and behaviors. *(*)*
Environmental server interaction	10%	We began investigating the data that will be used and potential interactions and behaviors.

Activity	*%*	*Observations*
Implement lane detection	100%	We have developed a lane detection algorithm using OpenCV.
Intersection detection	50%	We have been working on the detection of intersections and how to proceed when the car is at one.
Implement traffic sign detection	90%	The car is capable of recognizing traffic signs and reacting to most of them. The only signs for which the reaction has not yet been implemented are the *One-way Road* and *No-entry Road signs*; these will be included in the next report.
Implement traffic light detection	80%	We have started working and implementing the traffic light detector. Further testing is needed.
Object detection & classification	100%	The object detection pipeline is developed. We might retrain the model if necessary (if we add images to the dataset).
Architecture revision	100%	It was not necessary to modify our architecture.
Architecture diagram	100%	It was not necessary to modify our architecture diagrams.
Prepare traffic sign dataset	90%	The data set is almost ready, but we leave a percentage to improve it if necessary.
Define decision making	80%	We have improved the decision-making process. We have created a node map of our test racetrack to follow a path even in poor visibility scenarios, to improve the decision making.
Define robustness and safety measures	10%	We are working on the overall error handling and recovery of the system.
Document the 4th report	100%	The report itself.

(*) This activity is planned to last for two sprints to refine the car's reaction.

4 General status of project

The team has made significant progress in developing the car's algorithm. The car is already capable of following lanes, detecting and reacting to pedestrians and most of the traffic signs, adjusting its path, speed, or braking if necessary. The car can also know its position and follow a path represented as nodes in a graph, even when its vision is reduced, for example, driving in fog.

5 Upcoming activities

To be continued from the 4th sprint

- **Define other necessary sensors, define use-case, integration, preprocessing, noise cancelling.**
- **Define use-cases and test given servers information.**
- **Testing sensors and actuators for proper capture, processing and response.**
- **Environmental server interaction.**
- **Research hardware additions (headlight).**
- **Implement traffic sign detection.**
- **Intersection detection.**
- **Implement traffic light detection.**
- **Prepare traffic sign dataset.**
- **Define decision making (priorities of actions and state flow).**
- **Define robustness and safety measures.**

New activities

- **Introduce noise on systems to validate robustness (loss of image, burned image road search, undefined objects and states).**
- **Project presentation.**
- **Architecture revision**
- **Architecture diagram.**
- **Document the 5th report.**